Anomaly Cancellation in the Metric Bundle Pati-Salam Theory

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Abstract

We verify anomaly cancellation for the Pati-Salam gauge theory $\mathrm{SU}(4)\times\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$ that emerges from the metric bundle $Y^{14} = \mathrm{Met}(X^4)$. The fermion content derived in earlier work—the **16**-plet of $\mathrm{Spin}(10) \supset \mathrm{Spin}(6) \times \mathrm{Spin}(4)$, decomposing as $(\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$—is shown to be free of all gauge anomalies. The purely cubic $\mathrm{SU}(4)^3$ anomaly vanishes because $\mathbf{4}\oplus\bar{\mathbf{4}}$ is a real representation; all $\mathrm{SU}(2)^3$ anomalies vanish because the symmetric d-tensor of SU(2) is identically zero; and all mixed gauge-gravitational anomalies cancel because every gauge factor is non-abelian with traceless generators. The Witten SU(2) global anomaly is absent because each SU(2) factor sees four Weyl doublets (an even number), a consequence of the Pati-Salam unification of leptons with quarks as the fourth colour. After Pati-Salam breaking to $\mathrm{SU}(3)_c\times\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y$, all six Standard Model anomaly conditions are verified, including the cubic $\mathrm{U}(1)_Y^3$ and the mixed $\mathrm{U}(1)_Y$-gravitational anomaly.

Keywords: anomaly cancellation, Pati-Salam model, Spin(10) spinor, metric bundle, gauge anomalies

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1 Introduction

The metric bundle programme [1–3] derives the Pati-Salam gauge group and its fermion content from the geometry of $Y^{14} = \mathrm{Met}(X^4)$. The gauge group $\mathrm{SU}(4)\times\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$ arises as the maximal compact subgroup of $\mathrm{SO}(6,4)$ [1], and the fermion content is the **16**-plet of $\mathrm{Spin}(10) \supset \mathrm{Spin}(6) \times \mathrm{Spin}(4) \cong \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ [2].

A necessary condition for the quantum consistency of any chiral gauge theory is the cancellation of gauge anomalies [4, 5]. In four spacetime dimensions, the potentially dangerous anomalies are:

1. **Perturbative gauge anomalies**: From triangle diagrams with three gauge bosons, proportional to $\mathrm{Tr}[T^a\{T^b, T^c\}]$.
2. **Mixed gauge-gravitational anomalies**: From triangle diagrams with one gauge boson and two gravitons, proportional to $\mathrm{Tr}[T^a]$.
3. **Witten's global** SU(2) **anomaly** [6]: A non-perturbative anomaly present when an SU(2) gauge group has an odd number of Weyl doublets.

In this paper, we verify that all anomalies cancel for the metric bundle Pati-Salam theory. The cancellation is not a coincidence but a consequence of the Spin(10) origin of the fermion representation.

Outline. Section 2 reviews the fermion content. Section 3 verifies anomaly cancellation at the Pati-Salam level. Section 4 addresses the Witten global anomaly. Section 5 checks anomaly cancellation after breaking to the Standard Model. Section 6 discusses the pure gravitational anomaly and its relation to the generation number. Section 7 summarises.

2 Fermion Content

2.1 The Pati-Salam representation

The fermion content derived in [2] from the Clifford algebra of the positive-norm sector of the normal bundle is the **16**-plet of Spin(10):

$$\mathbf{16} = (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2}) \quad \text{under} \quad \mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R\,. \tag{1}$$

All fermions are left-handed Weyl spinors. The representation (1) has the following features:

- It is *vector-like* with respect to SU(4): the $\mathbf{4}$ and $\bar{\mathbf{4}}$ appear with equal multiplicity.
- It is *chiral* with respect to $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$: the $\mathbf{4}$ couples to $\mathrm{SU}(2)_L$ doublets and the $\bar{\mathbf{4}}$ to $\mathrm{SU}(2)_R$ doublets.
- It is left-right symmetric under the exchange $L \leftrightarrow R$ combined with $\mathbf{4} \leftrightarrow \bar{\mathbf{4}}$.

2.2 Standard Model decomposition

Under $\mathrm{SU}(4) \to \mathrm{SU}(3)_c \times \mathrm{U}(1)_{B-L}$, with hypercharge $Y = (B-L)/2 + T_{3R}$:

Field	$\mathrm{SU}(3)_c$	$\mathrm{SU}(2)_L$	Y	**Particle**
Q_L	$\mathbf{3}$	$\mathbf{2}$	$+\frac{1}{6}$	(u_L, d_L)
L_L	$\mathbf{1}$	$\mathbf{2}$	$-\frac{1}{2}$	(ν_L, e_L)
u_R^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$-\frac{2}{3}$	$\bar{u}_R$
d_R^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$+\frac{1}{3}$	$\bar{d}_R$
e_R^c	$\mathbf{1}$	$\mathbf{1}$	$+1$	$\bar{e}_R$
ν_R^c	$\mathbf{1}$	$\mathbf{1}$	0	$\bar{\nu}_R$

This is exactly one complete generation of Standard Model fermions plus a right-handed neutrino, with $6 + 2 + 3 + 3 + 1 + 1 = 16$ Weyl components.

3 Perturbative Gauge Anomalies at the Pati-Salam Level

The perturbative gauge anomaly in four dimensions is proportional to [4, 5]:

$$\mathcal{A}^{abc} = \sum_{\text{left Weyl}} \mathrm{Tr}\big[T^a\{T^b, T^c\}\big] \ , \tag{2}$$

summed over all left-handed Weyl fermions, where T^a are the gauge generators. For a product gauge group $G_1 \times G_2 \times G_3$, the possible anomaly types from triangle diagrams are G_i^3 (all three vertices from the same factor) and mixed types G_i^2-G_j or G_i-G_j-G_k. For non-abelian factors, the mixed types vanish: an anomaly of the form G_1^2-G_2 requires $\mathrm{Tr}_{G_2}[T^a]$ (a trace of a single generator), which is zero for any representation of a simple Lie group.

3.1 $\mathrm{SU}(4)^3$ anomaly

Proposition 3.1. $\mathcal{A}(\mathrm{SU}(4)^3) = 0$.

Proof. The cubic anomaly coefficient for a representation R of $\mathrm{SU}(N)$ is $A(R) = \mathrm{Tr}_R[T^a\{T^b, T^c\}]/d^{abc}_{\text{fund}}$, normalised so that $A(\text{fund}) = 1$. The fundamental property $A(\bar{R}) = -A(R)$ implies that the combined representation $R \oplus \bar{R}$ is anomaly-free.

The SU(4) content of the **16**-plet, with SU(2) multiplicities included, is:

$$\underbrace{2 \times \mathbf{4}}_{\text{from } (\mathbf{4},\mathbf{2},\mathbf{1})} \quad \oplus \quad \underbrace{2 \times \bar{\mathbf{4}}}_{\text{from } (\bar{\mathbf{4}},\mathbf{1},\mathbf{2})} \quad = 2(\mathbf{4} \oplus \bar{\mathbf{4}}) \, , \tag{3}$$

which is real. Therefore $\mathcal{A}(\mathrm{SU}(4)^3) = 2A(\mathbf{4}) + 2A(\bar{\mathbf{4}}) = 2A(\mathbf{4}) - 2A(\mathbf{4}) = 0$. □

3.2 $\mathrm{SU}(2)_L^3$ and $\mathrm{SU}(2)_R^3$ anomalies

Proposition 3.2. $\mathcal{A}(\mathrm{SU}(2)_L^3) = 0$ *and* $\mathcal{A}(\mathrm{SU}(2)_R^3) = 0$.

Proof. The totally symmetric tensor $d^{abc} = \mathrm{Tr}[T^a\{T^b, T^c\}]$ vanishes identically for SU(2), because SU(2) has only pseudo-real representations ($\bar{\mathbf{2}} \cong \mathbf{2}$ via the ϵ-tensor). Equivalently, the Dynkin index of the adjoint representation $A(\text{adj}) = 0$ for SU(2). Therefore the $\mathrm{SU}(2)^3$ anomaly is zero for *any* fermion content. □

3.3 Mixed gauge-gravitational anomalies

Proposition 3.3. *All mixed G^2-gravitational anomalies vanish at the Pati-Salam level.*

Proof. The mixed gauge-gravitational anomaly for a gauge group factor G_i is proportional to:

$$\mathcal{A}(G_i\text{-grav}^2) \propto \sum_{\text{left Weyl}} \mathrm{Tr}_{R_i}[T^a]\,. \tag{4}$$

For any representation R of a simple Lie group, $\mathrm{Tr}_R[T^a] = 0$ (the generators are traceless in every representation). Since all three Pati-Salam factors SU(4), $\mathrm{SU}(2)_L$, $\mathrm{SU}(2)_R$ are simple, this anomaly vanishes automatically. □

Remark 3.4. There is no U(1) factor at the Pati-Salam level, so no U(1)-gravitational anomaly needs to be checked. This type of anomaly becomes relevant only after Pati-Salam breaking; see Section 5.

3.4 Summary

Theorem 3.5 (Anomaly freedom at the Pati-Salam level)**.** *The Pati-Salam gauge theory* $\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ *with fermion content* $(\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$ *(left-handed Weyl) is free of all perturbative gauge anomalies and mixed gauge-gravitational anomalies.*

Anomaly type	**Mechanism**	**Status**
$\mathrm{SU}(4)^3$	$\mathbf{4} \oplus \bar{\mathbf{4}}$ is real	✓
$\mathrm{SU}(2)_L^3$	$d^{abc} = 0$ for SU(2)	✓
$\mathrm{SU}(2)_R^3$	$d^{abc} = 0$ for SU(2)	✓
G_i-grav^2	$\mathrm{Tr}_R[T^a] = 0$ (non-abelian)	✓

4 Witten's SU(2) Global Anomaly

Witten [6] showed that an SU(2) gauge theory with an odd number of Weyl doublets is inconsistent: the fermion path integral changes sign under a topologically non-trivial gauge transformation (exploiting $\pi_4(\mathrm{SU}(2)) \cong \mathbb{Z}_2$).

Proposition 4.1 (Absence of global anomaly)**.** *The metric bundle Pati-Salam theory is free of the Witten* SU(2) *anomaly for both* $\mathrm{SU}(2)_L$ *and* $\mathrm{SU}(2)_R$*.*

Proof. We count the number of left-handed Weyl doublets for each SU(2) factor:

$\mathrm{SU}(2)_L$ **doublets:** The $(\mathbf{4}, \mathbf{2}, \mathbf{1})$ contributes $\dim(\mathbf{4}) = 4$ doublets. The $(\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$ contributes zero doublets (all are $\mathrm{SU}(2)_L$ singlets). Total: 4 doublets (even).

$\mathrm{SU}(2)_R$ **doublets:** The $(\mathbf{4}, \mathbf{2}, \mathbf{1})$ contributes zero doublets (all are $\mathrm{SU}(2)_R$ singlets). The $(\bar{\mathbf{4}}, \mathbf{1}, \mathbf{2})$ contributes $\dim(\bar{\mathbf{4}}) = 4$ doublets. Total: 4 doublets (even).

Since both counts are even, neither SU(2) factor has a Witten anomaly. □

Remark 4.2 (The role of the fourth colour)**.** The four doublets per SU(2) arise from the dimension of the SU(4) fundamental. If the colour group were SU(3) instead of SU(4) (i.e., if leptons were not unified with quarks), each SU(2) would see 3 doublets, which is odd, and the Witten anomaly would render the theory inconsistent. The Pati-Salam structure—in which lepton number is the fourth colour [7]—is therefore essential for quantum consistency. This is a non-trivial check on the metric bundle's prediction of SU(4) rather than SU(3) × U(1).

5 Anomaly Cancellation after Standard Model Breaking

After the Pati-Salam breaking $\mathrm{SU}(4) \times \mathrm{SU}(2)_R \to \mathrm{SU}(3)_c \times \mathrm{U}(1)_Y$, the Standard Model gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ has six independent anomaly conditions [8]. We verify all of them for one generation.

The fermion content is (all left-handed Weyl): $Q_L = (\mathbf{3}, \mathbf{2})_{1/6}$ (6 components), $u_R^c = (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}$ (3 components), $d_R^c = (\bar{\mathbf{3}}, \mathbf{1})_{1/3}$ (3 components), $L_L = (\mathbf{1}, \mathbf{2})_{-1/2}$ (2 components), $e_R^c = (\mathbf{1}, \mathbf{1})_1$ (1 component), $\nu_R^c = (\mathbf{1}, \mathbf{1})_0$ (1 component).

5.1 $\mathrm{SU}(3)_c^3$

The SU(3) content is $2 \times \mathbf{3}$ (from Q_L) plus $\bar{\mathbf{3}} + \bar{\mathbf{3}}$ (from u_R^c and d_R^c):

$$2\,A(\mathbf{3}) + A(\bar{\mathbf{3}}) + A(\bar{\mathbf{3}}) = 2A(\mathbf{3}) - 2A(\mathbf{3}) = 0\,. \quad \checkmark \tag{5}$$

5.2 $\mathrm{SU}(2)_L^3$

Vanishes identically ($d^{abc} = 0$ for SU(2)). ✓

5.3 $\mathrm{SU}(3)_c^2$-$\mathrm{U}(1)_Y$

$$\sum_{\text{colour-charged}} Y_f \cdot T(\text{colour rep}) = 2 \cdot \tfrac{1}{6} \cdot \tfrac{1}{2} + 1 \cdot (-\tfrac{2}{3}) \cdot \tfrac{1}{2} + 1 \cdot \tfrac{1}{3} \cdot \tfrac{1}{2} = \tfrac{1}{6} - \tfrac{1}{3} + \tfrac{1}{6} = 0\,. \quad \checkmark \tag{6}$$

(The factor of 2 in the first term counts the $\mathrm{SU}(2)_L$ doublet multiplicity; $T(\mathbf{3}) = T(\bar{\mathbf{3}}) = 1/2$.)

5.4 $\mathrm{SU}(2)_L^2$-$\mathrm{U}(1)_Y$

$$\sum_{\text{doublets}} Y_f \cdot \dim(\text{colour rep}) \cdot T(\mathbf{2}) = 3 \cdot \tfrac{1}{6} \cdot \tfrac{1}{2} + 1 \cdot (-\tfrac{1}{2}) \cdot \tfrac{1}{2} = \tfrac{1}{4} - \tfrac{1}{4} = 0\,. \quad \checkmark \tag{7}$$

(Colour triplet Q_L contributes with multiplicity 3; colour singlet L_L contributes with multiplicity 1.)

5.5 $\mathrm{U}(1)_Y^3$

Summing Y_f^3 over all 16 Weyl components:

$$\begin{aligned}\sum_f Y_f^3 &= 6\cdot\left(\tfrac{1}{6}\right)^3 + 3\cdot\left(-\tfrac{2}{3}\right)^3 + 3\cdot\left(\tfrac{1}{3}\right)^3 + 2\cdot\left(-\tfrac{1}{2}\right)^3 + 1\cdot 1^3 + 1\cdot 0^3 \\ &= \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 \\ &= \tfrac{1}{36} - \tfrac{32}{36} + \tfrac{4}{36} - \tfrac{9}{36} + \tfrac{36}{36} = 0\,. \quad \checkmark \end{aligned} \tag{8}$$

5.6 $\mathrm{U}(1)_Y$-grav^2 (mixed gravitational)

$$\sum_f Y_f = 6\cdot\tfrac{1}{6} + 3\cdot(-\tfrac{2}{3}) + 3\cdot\tfrac{1}{3} + 2\cdot(-\tfrac{1}{2}) + 1 + 0 = 1 - 2 + 1 - 1 + 1 = 0\,. \quad \checkmark \tag{9}$$

5.7 Summary

Theorem 5.1 (SM anomaly cancellation)**.** *All six perturbative anomaly conditions of the Standard Model gauge group are satisfied by the fermion spectrum derived from the metric bundle* **16***-plet, for each generation.*

This reproduces the well-known result [8] that the Standard Model fermion content is anomaly-free. The non-trivial point is that the metric bundle *derives* this fermion content rather than assuming it.

6 The Pure Gravitational Anomaly and Generation Number

6.1 The gravitational anomaly condition

The pure gravitational anomaly in four dimensions is proportional to the difference $N_L - N_R$ between left-handed and right-handed Weyl fermions. For a single **16**-plet (all left-handed in our convention):

$$N_L - N_R = 16 \quad \text{(per generation)}\,. \tag{10}$$

In even dimensions $D = 4k + 2$, the gravitational anomaly cancellation on a compact spin manifold requires that $N_L - N_R$ satisfy a divisibility condition related to the $\hat{A}$-genus. In $D = 4$ on a compact spin manifold, the condition is [9]:

$$N_L - N_R \equiv 0 \pmod 1\,, \tag{11}$$

which is always satisfied. However, for modular invariance of the partition function on spin manifolds with torsion, a stronger condition is sometimes required:

$$N_L - N_R \equiv 0 \pmod{24}\,. \tag{12}$$

6.2 Three generations

For N_G generations of **16**-plets:

$$N_L - N_R = 16\, N_G\,. \tag{13}$$

Divisibility by 24 requires $16\, N_G \equiv 0 \pmod{24}$, i.e., $2\, N_G \equiv 0 \pmod 3$, which holds if and only if N_G is divisible by 3.

Remark 6.1. The smallest value $N_G = 3$ is distinguished by two conditions:

1. $16 \times 3 = 48 = 2 \times 24$, satisfying the modular constraint.
2. $N_G \geq 3$ is required for CP violation via the CKM phase [10].

While this does not explain why $N_G = 3$, it provides a non-trivial consistency check: the metric bundle framework, combined with the gravitational constraint, permits $N_G \in \{3, 6, 9, \ldots\}$, and the smallest such value matches observation.

6.3 The index theorem on Y^{14}

In the full fourteen-dimensional theory, the number of chiral zero modes is determined by the Atiyah-Singer index theorem:

$$\mathrm{ind}(\not{D}) = \int_{Y^{14}} \hat{A}(TY) \wedge \mathrm{ch}(V)\,, \tag{14}$$

where V is the gauge bundle. For the metric bundle $Y = \mathrm{Met}(X)$, the topology of Y is determined by the topology of X and the fibre $\mathrm{GL}^+(4,\mathbb{R})/\,\mathrm{SO}(4)$. Since the fibre is contractible (diffeomorphic to $\mathbb{R}^{10}$), the topology of Y is "inherited" from X, and the index depends on the topology of the base spacetime.

A detailed computation of $\mathrm{ind}(\not{D})$ for specific base topologies—and in particular whether it yields $N_G = 3$—is left for future work.

7 Discussion

We have verified that the fermion content of the metric bundle Pati-Salam theory is free of all gauge anomalies, both at the Pati-Salam and Standard Model levels. The results are summarised in the following table:

Anomaly	**Cancellation mechanism**	**Level**
$\mathrm{SU}(4)^3$	Real representation $(\mathbf{4} \oplus \bar{\mathbf{4}})$	PS
$\mathrm{SU}(2)^3_{L,R}$	$d^{abc} = 0$ for SU(2)	PS
G_i-grav^2	$\mathrm{Tr}_R[T^a] = 0$ (non-abelian)	PS
Witten SU(2)	4 doublets per SU(2) (even)	PS
$\mathrm{SU}(3)^3$	$2(\mathbf{3} \oplus \bar{\mathbf{3}})$	SM
$\mathrm{SU}(3)^2$- $\mathrm{U}(1)_Y$	Hypercharge sum cancels	SM
$\mathrm{SU}(2)^2$- $\mathrm{U}(1)_Y$	Hypercharge sum cancels	SM
$\mathrm{U}(1)^3_Y$	Cubic sum cancels	SM
$\mathrm{U}(1)_Y$-grav^2	Linear sum cancels	SM

Structural origin. The anomaly cancellation is ultimately guaranteed by the Spin(10) origin of the fermion representation. The **16** of Spin(10) is a chiral spinor of an orthogonal group, and chiral spinors of $\mathrm{Spin}(4k+2)$ are automatically anomaly-free for all subgroups [11]. The metric bundle provides the Spin(10) embedding via $\mathrm{Spin}(10) \supset \mathrm{Spin}(6) \times \mathrm{Spin}(4)$, where Spin(6) comes from the positive-norm sector and Spin(4) from the negative-norm sector of the DeWitt metric.

Open questions.

1. The fourteen-dimensional anomaly analysis, including the computation of $\hat{A}(TY)$ and the potential role of the Green-Schwarz mechanism [12], remains open.

2. The computation of $\mathrm{ind}(\not{D})$ on Y^{14} for specific base topologies may constrain or determine the generation number N_G.

3. The global structure of the gauge group (whether the theory sees $[\mathrm{SU}(4) \times \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R]/\mathbb{Z}_2$ or the simply connected cover) may impose additional constraints from discrete anomalies.

4. Paper VI [13] derives quantum mechanics—the Born rule, interference, and uncertainty—from the same $(6,4)$ DeWitt geometry, providing a deeper foundation for the quantum consistency of the metric bundle theory.

References

[1] Sloan Austermann. The Lorentzian DeWitt metric and the Pati-Salam gauge group. *Zenodo preprint*, 2026. Companion paper.

[2] Sloan Austermann. Fermion representations from the metric bundle: Clifford algebras and the Pati-Salam spinor. *Zenodo preprint*, 2026. Companion paper.

[3] Sloan Austermann. Gauge fields from the gauss equation of the metric bundle. *Zenodo preprint*, 2026. Companion paper.

[4] Stephen L. Adler. Axial-vector vertex in spinor electrodynamics. *Physical Review*, 177:2426–2438, 1969.

[5] David J. Gross and Roman Jackiw. Effect of anomalies on quasi-renormalizable theories. *Physical Review D*, 6:477–493, 1972.

[6] Edward Witten. An SU(2) anomaly. *Physics Letters B*, 117:324–328, 1982.

[7] Jogesh C. Pati and Abdus Salam. Lepton number as the fourth "color". *Physical Review D*, 10:275–289, 1974.

[8] Claude Bouchiat, John Iliopoulos, and Philippe Meyer. An anomaly-free version of Weinberg's model. *Physics Letters B*, 38:519–523, 1972.

[9] Luis Alvarez-Gaumé and Edward Witten. Gravitational anomalies. *Nuclear Physics B*, 234:269–330, 1984.

[10] Makoto Kobayashi and Toshihide Maskawa. CP-violation in the renormalizable theory of weak interaction. *Progress of Theoretical Physics*, 49:652–657, 1973.

[11] John C. Baez and John Huerta. The algebra of grand unified theories. *Bulletin of the American Mathematical Society*, 47:483–552, 2010.

[12] Michael B. Green and John H. Schwarz. Anomaly cancellations in supersymmetric $D = 10$ gauge theory and superstring theory. *Physics Letters B*, 149:117–122, 1984.

[13] Sloan Austermann. Quantum mechanics from finite observation: Born rule, interference, and uncertainty as consequences of Markov blanket geometry. *Zenodo preprint*, 2026. Companion paper.